

Modeling Likelihoods in Naive Bayes

Gaussian, Multinomial, Categorical, and Bernoulli Distributions

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Introduction: The Core of Naive Bayes

Naive Bayes methods are a set of supervised learning algorithms based on applying Bayes' theorem. Bayes' theorem defines the relationship for a class y and features x_1 through x_n :

$$P(y \mid x_1, \dots, x_n) = \frac{P(y)P(x_1, \dots, x_n \mid y)}{P(x_1, \dots, x_n)}$$

The “Naive” assumption simplifies this by assuming conditional independence between every pair of features given the class:

$$P(x_i \mid y, x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = P(x_i \mid y)$$

This yields our classification rule using Maximum A Posteriori (MAP) estimation:

MAP Decision Rule

$$\hat{y} = \arg \max_y P(y) \prod_{i=1}^n P(x_i \mid y)$$

Why Different Variants?

The core equation requires us to estimate the likelihood: $\mathbf{P}(\mathbf{x}_i | \mathbf{y})$. How we estimate this depends entirely on the nature of our data!

- **Continuous Data:** We assume a normal distribution $\rightarrow$ *Gaussian Naive Bayes*
- **Count/Frequency Data:** We look at counts of events (e.g., words) $\rightarrow$ *Multinomial Naive Bayes*
- **Categorical Data:** We look at distinct discrete categories $\rightarrow$ *Categorical Naive Bayes*
- **Binary/Boolean Data:** We look at presence/absence (0 or 1) $\rightarrow$ *Bernoulli Naive Bayes*

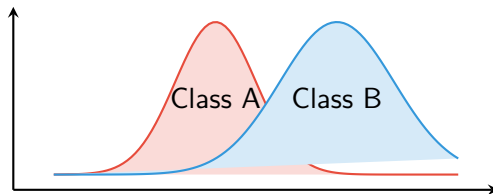
Let's dive into each variant!

Section 1

Gaussian Naive Bayes

Gaussian Naive Bayes: Primary Intuition

When our features are **continuous** (e.g., height, weight, temperature), we cannot simply count occurrences to find probabilities. **The Intuition:** We assume that the continuous values associated with each class are distributed according to a normal (Gaussian) distribution.



Instead of exact matches, we look at where a new data point falls on these curves.

Mathematical Formulation

Gaussian Naive Bayes assumes the likelihood of the features is Gaussian:

$$P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

Where:

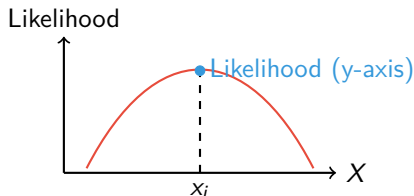
- x_i is the continuous feature value for a given sample.
- μ_y is the *mean* of feature i for class y .
- σ_y^2 is the *variance* of feature i for class y .

These parameters (μ_y and σ_y) are estimated from the training data using maximum likelihood (calculating the simple mean and standard deviation of the training set).

Crucial Concept: Likelihood vs. Probability

Wait, are we multiplying probabilities? Strictly speaking, the probability of a continuous random variable taking an *exact* value is **zero**. (e.g., the probability of being exactly 180.000000... cm tall is zero). Instead, we use **Likelihood**:

- The formula gives us the *y-axis coordinate* on the Gaussian curve.
- Likelihood values can actually be greater than 1 (unlike probabilities, which are bounded between 0 and 1).
- Because we are comparing relative scores across classes, comparing likelihoods works perfectly!



Numerical Example: The Setup

Let's predict if someone will like a new **Sci-Fi Movie**. We surveyed people based on two features: **Age** and **Books Read/Year**. **Training Data Summary:**

Likes Sci-Fi (Class: Yes) - Total 10 people

- Age: Mean $\mu = 25$, Std Dev $\sigma = 5$
- Books: Mean $\mu = 15$, Std Dev $\sigma = 4$

Does NOT Like (Class: No) - Total 10 people

- Age: Mean $\mu = 45$, Std Dev $\sigma = 10$
- Books: Mean $\mu = 4$, Std Dev $\sigma = 2$

New Person: Age = 30, Books = 12. **Priors:** $P(\text{Yes}) = 0.5$, $P(\text{No}) = 0.5$.

Numerical Example: Step-by-Step Execution

Step 1: Calculate Likelihoods for “Yes”

- Age (30) given Yes ($\mu = 25, \sigma = 5$): $L = \frac{1}{\sqrt{2\pi(25)}} \exp\left(-\frac{(30-25)^2}{2(25)}\right) \approx 0.048$
- Books (12) given Yes ($\mu = 15, \sigma = 4$): $L = \frac{1}{\sqrt{2\pi(16)}} \exp\left(-\frac{(12-15)^2}{2(16)}\right) \approx 0.075$

Step 2: Calculate Likelihoods for “No”

- Age (30) given No ($\mu = 45, \sigma = 10$): $L = \frac{1}{\sqrt{2\pi(100)}} \exp\left(-\frac{(30-45)^2}{2(100)}\right) \approx 0.013$
- Books (12) given No ($\mu = 4, \sigma = 2$): $L = \frac{1}{\sqrt{2\pi(4)}} \exp\left(-\frac{(12-4)^2}{2(4)}\right) \approx 0.000067$

Numerical Example: Final Scores

Step 3: Multiply Prior $\times$ Likelihoods Score for Yes:

$$Score(Yes) = P(Yes) \times L(Age|Yes) \times L(Books|Yes)$$

$$Score(Yes) = 0.5 \times 0.048 \times 0.075 = \mathbf{0.0018}$$

Score for No:

$$Score(No) = P(No) \times L(Age|No) \times L(Books|No)$$

$$Score(No) = 0.5 \times 0.013 \times 0.000067 = \mathbf{0.000000435}$$

Conclusion: Since $0.0018 \gg 0.000000435$, we classify the new person as **Yes (Likes Sci-Fi)**.

The Underflow Problem and the Log Trick

Did you notice how small our scores were getting? In real datasets with hundreds of features, multiplying hundreds of likelihoods (e.g., $0.06 \times 0.004 \times 0.000001 \dots$) yields a microscopic number. **The Problem:** Computers have a limit to how close a number can get to zero. When numbers get too small, the computer rounds them to exactly 0.0. This is called **Underflow**. **The Solution:** Logarithms! We take the natural logarithm ($\ln$) of the MAP equation.

$$\log(a \times b) = \log(a) + \log(b)$$

New Equation:

$$\hat{y} = \arg \max_y \left[\log(P(y)) + \sum_{i=1}^n \log(P(x_i | y)) \right]$$

This turns microscopic decimal multiplication into safe, negative number addition!

Gaussian NB Caveats: When Normality Fails

Gaussian NB assumes each continuous feature is normally distributed *within* each class. **If the feature is not normal, you have options:**

- **Transform the feature:** Apply log, square-root, Box–Cox, or Yeo–Johnson transforms to reduce skewness and stabilize variance.
- **Discretize / bin:** Convert continuous values into categories and use *Categorical* NB (e.g., quantile or equal-width bins).
- **Use a different likelihood model:** Fit a non-Gaussian distribution (e.g., log-normal, gamma) or a mixture model per class.
- **Nonparametric density:** Estimate $P(x_i | y)$ with kernel density estimation (KDE) and plug it into the NB rule.
- **Switch variants or models:** If data are counts or binary, use Multinomial/Bernoulli NB; otherwise consider a more flexible classifier.

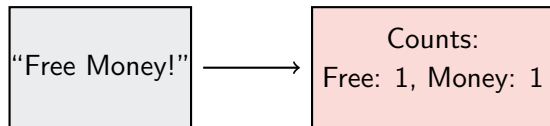
Section 2

Multinomial Naive Bayes

Multinomial Naive Bayes: Primary Intuition

When data can be represented as **counts** or **frequencies** (like word counts in a document), we use Multinomial Naive Bayes. **The Intuition:** We treat a text document as a “Bag of Words”.

- We ignore grammar and word order (this is why it's *Naive!*).
- We simply count how many times each word appears.
- We calculate the probability of seeing a specific word given the document class (e.g., Spam vs. Normal).



Mathematical Formulation

The distribution is parameterized by vectors $\theta_y = (\theta_{y1}, \dots, \theta_{yn})$ for each class y , where n is the vocabulary size. θ_{yi} is the probability $P(x_i | y)$ of feature (word) i appearing in class y . It is estimated by relative frequency:

$$\hat{\theta}_{yi} = \frac{N_{yi}}{N_y}$$

Where:

- N_{yi} : Number of times word i appears in class y .
- N_y : Total count of all words in class y .

The Zero Probability Problem

Imagine a new email arrives: “Lunch Money”. Let’s say in our training data, the word “Lunch” **never** appeared in a Spam email. Therefore, $P(\text{Lunch} \mid \text{Spam}) = 0$. When we calculate the score:

$$\text{Score}(\text{Spam}) = P(\text{Spam}) \times P(\text{Lunch} \mid \text{Spam}) \times P(\text{Money} \mid \text{Spam})$$

$$\text{Score}(\text{Spam}) = 0.33 \times \mathbf{0} \times 0.25 = \mathbf{0}$$

Because of one unseen word, the entire probability becomes ZERO, overriding the highly suspicious word “Money”. This breaks our model!

The Solution: Smoothing

To fix this, we add a “fake count” (smoothing prior $\alpha \geq 0$) to every word in our vocabulary.

$$\hat{\theta}_{yi} = \frac{N_{yi} + \alpha}{N_y + \alpha n}$$

Where:

- α is the smoothing parameter.
- n is the total number of unique words (vocabulary size).

Types of Smoothing:

- **Laplace Smoothing:** Setting $\alpha = 1$. (Adding exactly 1 count to everything).
- **Lidstone Smoothing:** Setting $0 < \alpha < 1$. (Adding a fractional count).

Spam/Ham Numerical Example: The Setup

Training Data Vocab Size (n) = 5 unique words (Dear, Friend, Lunch, Money, Hello).

Priors: Normal (Ham) = $2/3$, Spam = $1/3$. **Word Counts in Training:**

- **Normal (Total Words $N_{Normal} = 17$):**
Dear (8), Friend (5), Lunch (3), Money (1).
- **Spam (Total Words $N_{Spam} = 7$):**
Dear (2), Friend (1), Lunch (0), Money (4).

New Message to Classify: “Dear Friend”

Spam/Ham Numerical Example: Without Smoothing

Calculating for Normal (Ham):

- $P(\text{Dear}|\text{Normal}) = \frac{8}{17} \approx 0.47$
- $P(\text{Friend}|\text{Normal}) = \frac{5}{17} \approx 0.29$
- $\text{Score}(\text{Normal}) = P(\text{Normal}) \times 0.47 \times 0.29$
- $\text{Score}(\text{Normal}) = 0.67 \times 0.47 \times 0.29 \approx \mathbf{0.091}$

Calculating for Spam:

- $P(\text{Dear}|\text{Spam}) = \frac{2}{7} \approx 0.29$
- $P(\text{Friend}|\text{Spam}) = \frac{1}{7} \approx 0.14$
- $\text{Score}(\text{Spam}) = P(\text{Spam}) \times 0.29 \times 0.14$
- $\text{Score}(\text{Spam}) = 0.33 \times 0.29 \times 0.14 \approx \mathbf{0.013}$

Conclusion: $0.091 > 0.013 \rightarrow$ Classify as **Normal**.

Spam/Ham Numerical Example: With Laplace Smoothing

Now, let's classify "Lunch Money" using Laplace Smoothing ($\alpha = 1, n = 5$). **Formulas:**

$$P(\text{word}|\text{class}) = \frac{\text{Count}+1}{\text{Total}+5} \quad \text{Calculating for Normal:}$$

- $P(\text{Lunch}|\text{Normal}) = \frac{3+1}{17+5} = \frac{4}{22} \approx 0.18$
- $P(\text{Money}|\text{Normal}) = \frac{1+1}{17+5} = \frac{2}{22} \approx 0.09$
- $\text{Score}(\text{Normal}) = 0.67 \times 0.18 \times 0.09 \approx \mathbf{0.0108}$

Calculating for Spam (Notice how zero is fixed!):

- $P(\text{Lunch}|\text{Spam}) = \frac{0+1}{7+5} = \frac{1}{12} \approx 0.083$
- $P(\text{Money}|\text{Spam}) = \frac{4+1}{7+5} = \frac{5}{12} \approx 0.417$
- $\text{Score}(\text{Spam}) = 0.33 \times 0.083 \times 0.417 \approx \mathbf{0.0114}$

Conclusion: $0.0114 > 0.0108 \rightarrow$ Classify as **Spam**.

Section 3

Categorical Naive Bayes

Categorical Naive Bayes: Primary Intuition

What if our data isn't text counts or continuous numbers, but **discrete categories**?

Examples:

- Weather Outlook: Sunny, Overcast, Rain
- Wind: Weak, Strong
- Education Level: High School, BS, MS, PhD

Categorical Naive Bayes assumes each feature has its own distinct categorical distribution. It assumes data is encoded numerically (e.g., Sunny=0, Overcast=1, Rain=2).

Mathematical Formulation

For each feature i , the probability of category t given class c is:

$$P(x_i = t \mid y = c; \alpha) = \frac{N_{tic} + \alpha}{N_c + \alpha n_i}$$

Where:

- N_{tic} : Number of times category t appears in feature i for class c .
- N_c : Total number of samples belonging to class c .
- n_i : Number of available categories for feature i (e.g., 3 for weather).
- α : Smoothing parameter.

Zero Probability & Smoothing in Categorical Data

Just like in text, we can have the Zero Probability Problem. If in our training data, it **never rained** when we decided to “Play Tennis”, $P(Rain|Yes) = 0$. Any future prediction where the weather is “Rain” will automatically result in a 0 probability for “Play Tennis”, ignoring all other positive features. We apply **Laplace** ($\alpha = 1$) or **Lidstone** ($\alpha < 1$) smoothing in the exact same way. It adds a pseudo-count to N_{tic} and scales the denominator by the number of categories n_i .

Numerical Example: Play Tennis Setup

Goal: Predict if we Play Tennis (Yes/No) based on Outlook and Wind. **Priors:**

$P(Yes) = \frac{9}{14} \approx 0.64$, $P(No) = \frac{5}{14} \approx 0.36$. **Feature Counts from 14 Days:**

- **Outlook Categories ($n_1 = 3$):**

- Yes (9 total): Sunny (2), Overcast (4), Rain (3)
- No (5 total): Sunny (3), Overcast (0), Rain (2)

- **Wind Categories ($n_2 = 2$):**

- Yes (9 total): Weak (6), Strong (3)
- No (5 total): Weak (2), Strong (3)

New Instance: Outlook = Overcast, Wind = Strong.

Let's calculate with Laplace smoothing ($\alpha = 1$).

Numerical Example: Step-by-step Calculation

Step 1: Calculate Probabilities for Class “Yes” ($N_{Yes} = 9$)

- $P(Overcast|Yes) = \frac{4+1}{9+(1 \times 3)} = \frac{5}{12} \approx 0.417$
- $P(Strong|Yes) = \frac{3+1}{9+(1 \times 2)} = \frac{4}{11} \approx 0.364$
- $Score(Yes) = P(Yes) \times P(Overcast|Yes) \times P(Strong|Yes)$
- $Score(Yes) = 0.64 \times 0.417 \times 0.364 \approx \mathbf{0.097}$

Step 2: Calculate Probabilities for Class “No” ($N_{No} = 5$)

- $P(Overcast|No) = \frac{0+1}{5+(1 \times 3)} = \frac{1}{8} \approx 0.125$
- $P(Strong|No) = \frac{3+1}{5+(1 \times 2)} = \frac{4}{7} \approx 0.571$
- $Score(No) = P(No) \times P(Overcast|No) \times P(Strong|No)$
- $Score(No) = 0.36 \times 0.125 \times 0.571 \approx \mathbf{0.026}$

Conclusion: $0.097 > 0.026 \rightarrow \text{Play Tennis (Yes)}.$

Section 4

Bernoulli Naive Bayes

Bernoulli Naive Bayes: Primary Intuition

Bernoulli Naive Bayes is designed for data that is distributed according to multivariate Bernoulli distributions. **Intuition:** Every feature must be a **binary/boolean** variable (0 or 1).

- Did the word “Free” appear in the email? (Yes=1, No=0)
- We do not care if it appeared 1 time or 100 times. We only care *if* it appeared.

How it differs from Categorical NB:

- It is a special case of Categorical NB where $n_i = 2$ for all features.
- However, it uses a specific formulation that **explicitly penalizes the non-occurrence** of a feature.
- Multinomial NB ignores words that don't appear in the document. Bernoulli explicitly factors in the absence of a word.

Mathematical Formulation

The decision rule explicitly evaluates both presence ($x_i = 1$) and absence ($x_i = 0$):

$$P(x_i | y) = P(x_i = 1 | y)x_i + (1 - P(x_i = 1 | y))(1 - x_i)$$

Since $x_i \in \{0, 1\}$, this equation acts as an “if-else” statement:

- If $x_i = 1$, the formula returns $P(x_i = 1 | y)$.
- If $x_i = 0$, the formula returns $1 - P(x_i = 1 | y)$, representing the probability of the feature *not* occurring.

Numerical Example: Setup

Let's classify an article as **Sports** (Yes) or **Not Sports** (No) based on the presence of 3 words: "Ball", "Win", "Tax". **Priors:** $P(Sports) = 0.5$, $P(NotSports) = 0.5$. **Probabilities of**

Word Presence ($P(x_i = 1|y)$) **estimated from training:**

- $P(Ball = 1|Sports) = 0.8 \rightarrow P(Ball = 0|Sports) = 0.2$
- $P(Win = 1|Sports) = 0.6 \rightarrow P(Win = 0|Sports) = 0.4$
- $P(Tax = 1|Sports) = 0.1 \rightarrow P(Tax = 0|Sports) = 0.9$

New Document: Contains "Ball", but **NOT** "Win", **NOT** "Tax".

Vector: $x = [1]$.

Numerical Example: Step-by-step Calculation

Let's calculate the score for **Sports**: **Formula Application**:

$$\begin{aligned} \text{Score}(\text{Sports}) &= P(\text{Sports}) \\ &\times P(\text{Ball} = 1 | \text{Sports}) \quad (\text{because } x_1 = 1) \\ &\times (1 - P(\text{Win} = 1 | \text{Sports})) \quad (\text{because } x_2 = 0) \\ &\times (1 - P(\text{Tax} = 1 | \text{Sports})) \quad (\text{because } x_3 = 0) \end{aligned}$$

Plugging in the numbers:

$$\begin{aligned} \text{Score}(\text{Sports}) &= 0.5 \times 0.8 \times 0.4 \times 0.9 \\ \text{Score}(\text{Sports}) &= \mathbf{0.144} \end{aligned}$$

Notice how the *absence* of the word “Win” explicitly lowered the probability (multiplied by 0.4 instead of ignoring it)!

Summary & Conclusion

Choosing the Right Variant

- **Gaussian:** Continuous data (age, height, temperature). Evaluates using the normal distribution curve (Likelihood).
- **Multinomial:** Count data (word frequencies in a document). Evaluates using relative frequencies.
- **Categorical:** Discrete categorical data (weather types, colors). Evaluates relative frequencies across categories.
- **Bernoulli:** Binary data (word presence/absence). Specifically penalizes non-occurrences of features.

Universal Fixes:

- **Logarithms:** Prevents computer underflow when multiplying many small likelihoods.
- **Smoothing (α):** Prevents the Zero Probability Problem by adding baseline counts.

Thank You